
CLINICAL INVESTIGATION

Ruthenium-106 Plaque Radiotherapy Alone or in Combination with Transpupillary Thermotherapy in the Management of Choroidal Melanoma

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Abstract

Purpose: To evaluate the results of ruthenium-106 (Ru-106) plaque radiotherapy alone (group A) or in combination with transpupillary thermotherapy (TTT) (group B) in the management of choroidal melanoma with tumor thickness (height) <8 mm. The tumors in each group were subclassified as those with thickness ≤5 mm versus those with thickness of >5 and <8 mm.

Methods: In this retrospective review, the main outcome measures were globe conservation rate, the rate of a reduction of at least 50% in tumor thickness, treatment complications, visual acuity (VA) change, and metastasis. Kaplan-Meier curves for prediction of decrease in tumor thickness of at least 50% over time were constructed.

Results: A total of 54 patients (24 in group A and 30 in group B) were included in this study. The groups were matched with respect to patient age, tumor base diameter, tumor thickness, tumor distance to optic disc, tumor distance to foveola, and baseline visual acuity (VA). The mean follow-up was 24.6 months in group A and 44.9 months in group B. Globe conservation was achieved in 21 (87.5%) eyes in group A and in 26 (86.7%) eyes in group B. The globe conservation rates did not differ significantly between groups A and B or between tumors ≤5 mm in thickness and those >5 to <8 mm in thickness in each group ($P > 0.05$). There was no statistical difference between groups A and B in the rate of tumor thickness reduction of at least 50% ($P > 0.05$). There was a significant decrease in final VA compared to baseline VA in group B ($P = 0.007$) but not in group A. Radiation complications were similar in groups A and B. Liver metastasis occurred in two patients in group A and in one patient in group B. Statistical analysis could not be carried out for the latter two variables because of the small number of affected patients.

Conclusions: Compared to Ru-106 plaque radiotherapy alone, Ru-106 plaque radiotherapy combined with TTT did not result in a significant change in the globe conservation rate or the rate of at least 50% reduction in tumor thickness in choroidal melanomas <8 mm in thickness. Although Ru-106 plaque radiotherapy is mainly used for choroidal melanomas ≤5 mm thick, it can also be considered in selected tumors with thickness between 5 and 8 mm with comparable tumor control. **Jpn J Ophthalmol** 2010;54:338–343 © Japanese Ophthalmological Society 2010

Keywords: choroidal melanoma, iodine-125 plaque radiotherapy, ruthenium-106 plaque radiotherapy, transpupillary thermotherapy

Introduction

Plaque radiotherapy is one of the most common treatment modalities for choroidal and ciliary body melanoma. The

Collaborative Ocular Melanoma Study (COMS) found no significant difference between enucleation and iodine-125 plaque radiotherapy for medium-sized choroidal melanomas with respect to patient survival.¹ Therefore, eye-conserving treatment methods, including plaque radiotherapy, are gaining popularity.

Many radioactive isotopes, such as cobalt-60 (Co-60), ruthenium-106 (Ru-106), iodine-125 (I-125), iridium-192

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(Ir-192), and palladium-103 (Pd-103), have been used in ocular brachytherapy.² In 1959, Stallard³ reported using Co-60 radioisotope to treat uveal melanoma. Although the high-energy Co-60 radioisotope has deep tissue penetration and provided good tumor control, it is associated with high complication rates.⁴ In 1964, Lommatzsch et al.⁵ reported on the use of Ru-106 plaque radiotherapy, which provided lower energy and lower radiation complication rates than did Co-60, to treat uveal melanoma. Iodine-125 radioactive plaques also became popular as a treatment for uveal melanoma, providing lower energy and lower complication rates than Co-60.⁶

The half-life of Ru-106 is 369 days, which allows a plaque to be used successively on a number of patients, reducing the per-patient cost and making it affordable in many developing countries. Ru-106 emits beta radiation, which, because of its limited penetration, can be used to treat tumors up to 5 mm in thickness (height). Because I-125 has a half-life of only 60 days, it is available for use in a smaller number of patients than the longer-lived Ru-106. Iodine-125 emits gamma rays and can be used to treat tumors up to 10 mm in thickness.⁷ Ru-106 plaque radiotherapy seems to offer tumor control rates similar to those achieved with I-125 plaque radiotherapy.⁴ However, some studies have reported a lower tumor control rate with Ru-106 plaque radiotherapy.⁸

Ru-106 plaque radiotherapy as a monotherapy is usually recommended for tumors up to 5 mm in thickness. However, we used this treatment in tumors up to 8 mm in thickness on the basis of previously published studies.^{4,9} Hungerford⁴ and Bergman et al.⁹ indicated that Ru-106 plaque radiotherapy can be used to treat tumors >5 mm in thickness, delivering 120 Gy to a 5-mm tumor depth. The tumor apex receives a <85-Gy radiation dose, which is the normally recommended dose for uveal melanoma. The rationale for this application may be to close tumor vessels by delivering high-dose radiotherapy to the tumor base. Bergman et al.⁹ recommended that Ru-106 plaque radiotherapy should probably be reserved for ciliary body and choroidal melanomas <7 mm in thickness.

Transpupillary thermotherapy (TTT) using an infrared diode laser was initially introduced as a method to treat selected choroidal melanomas in conjunction with plaque radiotherapy. Experimental studies showed that thermal penetration reaches a maximum depth of 4 mm.¹⁰ On the basis of the information learned from experimental studies, TTT has been employed as primary treatment in small posteriorly located choroidal melanomas less than 4 mm in thickness.^{11,12} However, primary TTT has high complication and tumor recurrence rates, and primary TTT is seldom used today. The combination of TTT with brachytherapy is called “sandwich therapy,” a term reflecting the combined efficacy of brachytherapy at the base of the tumor with the efficacy of TTT at the apex.¹³ This approach may provide an opportunity to treat thick choroidal melanomas that would be unsuitable for treatment with brachytherapy alone. Other researchers also pointed out that combining brachytherapy with TTT is preferred for better local tumor control

in posteriorly located choroidal melanomas close to the optic disc and foveola.¹⁴

The purpose of this study was to evaluate the results of Ru-106 plaque radiotherapy alone and in combination with TTT in the management of choroidal melanoma with tumor thickness up to 8 mm. Our primary goals were to see if additional TTT provided any advantage over Ru-106 plaque radiotherapy alone in the treatment of choroidal melanoma in terms of globe conservation rate, at least 50% reduction in tumor thickness rate, treatment complications, visual acuity (VA) change, and metastasis. The differences in globe conservation rate and treatment complications were also analyzed between tumors ≤5 mm in thickness versus those from >5 to <8 mm in thickness in each group.

Materials and Methods

We reviewed the clinical charts of 54 patients with choroidal melanoma treated with Ru-106 plaque radiotherapy at the Ocular Oncology Service of Ankara University's Faculty of Medicine between July 2000 and September 2005. An Institutional Review Board approval was obtained. The recruited patients were followed up until September 2008. Each patient underwent complete ophthalmologic examination, including VA testing, intraocular pressure measurement, biomicroscopy, indirect ophthalmoscopy, A- and B-scan ultrasonography, and fluorescein angiography. Liver function tests, chest X-ray, and abdominal ultrasonography were performed in all patients for systemic evaluation.

Ru-106 plaque radiotherapy was performed either alone (group A) or in combination with TTT (group B). The groups were matched with respect to patient age, tumor base diameter, tumor thickness, tumor distance to optic disc, tumor distance to foveola, and baseline VA. The tumors of groups A and B were subclassified into those ≤5 mm in thickness (height) versus those >5 mm to <8 mm in thickness. In both groups, choroidal melanomas ≤5 mm in thickness were treated with an apex dose of 85–100 Gy. For tumors >5 mm in thickness, we aimed to give a dose of 120 Gy to a tumor depth of 5 mm, in accordance with a previous study.⁴ In these cases, the tumor apex received <85 Gy. In group B, the first TTT session was provided at the time of plaque removal and subsequent TTT applications were provided at 3-month intervals.

We used either notched (20 mm in diameter) or round plaques (15 or 20 mm in diameter). Notched plaques were used for juxtapapillary choroidal melanomas. Radiation calculations were made using the Bebig plaque simulator program (Bebig, Berlin, Germany). Patients were followed up at 1 month and 3 months after Ru-106 plaque radiotherapy and at 3- to 6-month intervals thereafter.

At each follow-up visit, complete ocular examination and A- and B-mode ultrasonography were performed. Fluorescein angiographic studies were repeated as necessary. Changes in tumor thickness, the local recurrence rate, the globe conservation rate, complications, and changes in VA

were recorded. Liver function tests, chest X-ray, and abdominal ultrasonography were performed at 6-month intervals to detect systemic metastasis.

The main outcome measures, the globe conservation rate, the rate of a reduction of at least 50% in tumor thickness, treatment complications, VA change, and metastasis, were compared between groups A and B. VA at baseline and final follow-up were compared in each group. For subgroup analysis of tumors ≤ 5 mm in thickness versus those >5 to <8 mm in thickness in each group, the differences in globe conservation rate and treatment complications were analyzed.

For statistical analysis, the *t* test for independent samples was used for continuous variables, including age, tumor base diameter, and tumor thickness. The Mann-Whitney *U* test was used for continuous variables, including distance to optic disc and distance to foveola. For comparison of globe conservation rates, the Fisher exact test was used. The change in tumor thickness during follow-up in each group was analyzed using the log-rank test. Kaplan-Meier curves for prediction of a rate of decrease of at least 50% in tumor thickness over time were constructed for both the Ru-106 plaque radiotherapy and the combined Ru-106 plaque radiotherapy and TTT groups. The difference between groups A and B of at least 50% decrease in tumor thickness rate was analyzed using the log rank test. The difference between preoperative and postoperative VAs, as logarithm of the minimum angle of resolution (logMAR) values, in each group was analyzed using the Mann-Whitney *U* test. The difference in preoperative VA between groups A and B was analyzed using the Wilcoxon test.

Results

Group A

Group A comprised 24 patients (12 women, 12 men) treated with Ru-106 plaque radiotherapy alone. The mean age of the patients was 57.9 years (range, 26.0–85.0 years). Of the 24 patients, 18 had no systemic disease, two had diabetes mellitus, two had systemic hypertension, and two had both diabetes mellitus and systemic hypertension. VA at baseline on the logMAR scale was 0.99.

The mean largest tumor basal diameter was 9.3 mm (median, 9.0 mm; range, 4.0–15.0 mm), and the mean initial tumor thickness was 5.3 mm (median, 4.8 mm; range, 2.6–8.0 mm). The mean distance to the optic disc was 4.0 mm (median, 1.5 mm; range, 0–8.0 mm), and to the foveola it was 3.0 mm (median, 3.0 mm; range, 0–10.0 mm). Of the 24 eyes, 14 (58.3%) had subretinal fluid initially.

Round plaques were used in 13 eyes (54.2%), and notched plaques were used in 11 (45.8%). The mean tumor apex radiation dose was 88.3 Gy (median, 85.5 Gy; range, 61.9–120.0 Gy), and the mean tumor base radiation dose was 629.6 Gy (median, 585.2 Gy; range, 370–1858 Gy). The mean tumor apex radiation dose rate was 57.9 cGy/h (median, 56.3 cGy/h, range, 20.5–90.3 cGy/h), and the mean

tumor base radiation dose rate was 384.6 cGy/h (median, 352.5 cGy/h; range, 192.4–585.4 cGy/h).

Mean tumor thickness decreased significantly from 5.3 mm (range, 2.6–8.0 mm) to 3.0 mm (range, 0–7.0 mm) during a mean follow-up of 24.6 months (range, 4.0–60.0 months). Mean tumor thickness was 3.7 mm at 3 months, 3.1 mm at 6 months, 2.8 mm at 12 months, 2.6 mm at 18 months, 3.0 mm at 24 months, 2.5 mm at 30 months, 2.1 mm at 36 months, 2.3 mm at 42 months, 2.0 mm at 48 months, 2.0 mm at 54 months, and 2.5 mm at 60 months (Fig. 1). The decrease in tumor thickness was significant compared to baseline at 6 months and thereafter (log-rank test, $P < 0.015$). Kaplan-Meier curves predicted at least a 50% decrease in tumor thickness in 92% of the eyes in group A by 48 months after treatment (Fig. 2).

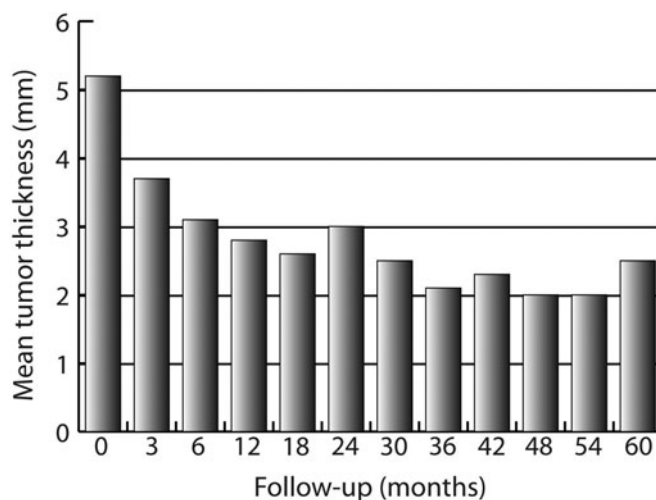


Figure 1. Mean tumor thickness decrease in patients treated with Ru-106 plaque radiotherapy alone (group A).

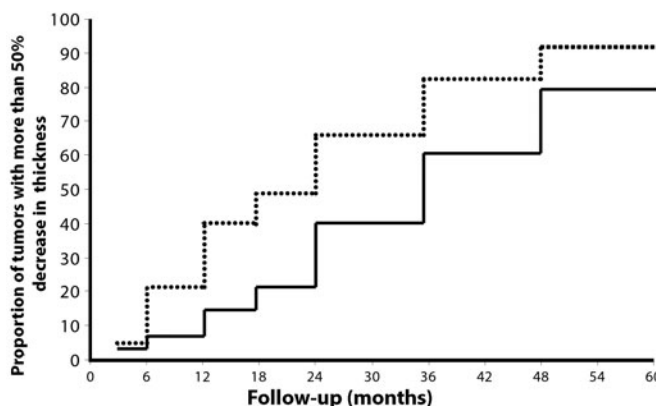


Figure 2. Kaplan-Meier estimates of an at least 50% reduction in tumor thickness in 24 patients treated with Ru-106 plaque radiotherapy and in 30 patients treated with combined Ru-106 plaque radiotherapy and transpupillary thermotherapy (TTT). , group A; ———, group B.

The regressed tumors became more reflective and more acoustically solid on A- and B-mode ultrasonography, respectively. On fluorescein angiography, the regressed tumors demonstrated decreased fluorescence in the late phases of the angiogram.

Complications in group A included radiation retinopathy in five eyes (20.8%), vitreous hemorrhage in five (20.8%), cataract in two (8.3%), pigment dispersion into the vitreous cavity in two (8.3%), increase in exudative retinal detachment in two (8.3%), scleral thinning in one (4.1%), and radiation papillopathy in one eye (4.1%). None of the eyes with radiation retinopathy required laser photocoagulation. Phacoemulsification surgery with intraocular lens implantation was performed in one eye with dense cataract. Eyes with increase in exudative retinal detachment after Ru-106 plaque radiotherapy were treated with a tapering dose of 1 mg/kg per day of oral prednisolone. Exudative retinal detachment resolved completely following treatment.

We analyzed the complication rates of tumors with thickness ≤ 5 mm (15 eyes) compared to those >5 to <8 mm in thickness (nine eyes). Complications that occurred in the 15 eyes with tumors ≤ 5 mm in thickness were vitreous hemorrhage in three eyes (20%), radiation retinopathy in two (13.3%), and cataract, radiation papillopathy, exudative retinal detachment, and pigment dispersion into the vitreous cavity in one eye (6.6%) each. Complications seen in the nine eyes with tumors between >5 and <8 mm in thickness were radiation retinopathy in three eyes (33.3%), vitreous hemorrhage in two (22.2%), and cataract, pigment dispersion into the vitreous cavity, scleral thinning, and exudative retinal detachment in one eye (11.1%) each. Eyes having tumors >5 to <8 mm in thickness had a higher rate of vitreous hemorrhage and cataract compared to eyes with tumors ≤ 5 mm in thickness. Because of the small number of patients in each complication group, a statistical analysis could not be carried out.

Globe conservation was achieved in 13 of 15 eyes (86.7%) with tumors ≤ 5 mm in thickness and in eight of nine eyes (88.9%) with tumors >5 to <8 mm in thickness. There was no significant difference between these groups with respect to the globe conservation rate (Fisher exact test, $P > 0.05$).

Excluding the three eyes that were enucleated, three of 21 eyes (14.2%) had VA $\geq 20/40$ initially. At the mean follow-up of 24.6 months (range, 4.0–60.0 months), three of 21 eyes (14.2%) maintained VA $\geq 20/40$. Initially, 13 of 21 eyes (61.9%) had VA $\leq 20/200$, and at the final follow-up 11 of 21 eyes (52.4%) had VA $\leq 20/200$. Final VA on the logMAR scale was 1.04 (excluding the eyes that were enucleated). There was no significant change between preoperative and postoperative logMAR VA (Mann-Whitney U test, $P > 0.05$).

Tumor recurred in two eyes (8.3%). Enucleation was performed in three eyes (12.5%) (for tumor recurrence in two eyes and for nonresolving vitreous hemorrhage in one eye). Liver metastasis developed in two patients at the 23 and 38 month follow-up, respectively, after plaque treatment.

Group B

Group B consisted of 30 patients (17 women, 13 men) treated with combined Ru-106 plaque radiotherapy and TTT. The mean age of the patients was 53.1 years (range, 26.0–73.0 years). Of the 30 patients, 24 had no systemic disease, one had diabetes mellitus, three had systemic hypertension, and two had both diabetes mellitus and systemic hypertension. Visual acuity at baseline on the logMAR scale was 0.9. The mean largest tumor basal diameter was 9.6 mm (median, 10.0 mm; range, 6.0–14.0 mm) and the mean initial tumor thickness was 5.2 mm (median, 5.0 mm; range, 2.0–8.0 mm). The mean distance to the optic disc was 1.5 mm (median, 1.0 mm; range, 0–4.0 mm), and to the foveola it was 1.9 mm (median, 1.0 mm; range, 0–6.0 mm). Of the 30 eyes, 26 (86.7%) had subretinal fluid initially.

Round plaques were used in 14 eyes (46.7%) and notched plaques were used in the other 16 eyes (53.3%). The mean tumor apex radiation dose was 94.5 Gy (median, 93.5 Gy; range, 75.0–120.0 Gy), and the mean tumor base radiation dose was 665.8 Gy (median, 628.2 Gy; range, 229.5–2070.0 Gy). The mean tumor apex radiation dose rate was 68.6 cGy/hr (median, 57.5 cGy/h; range, 25.7–167.0 cGy/h), and the mean tumor base radiation dose rate was 431.3 cGy/h (median, 415.1 cGy/h; range, 225.1–848.9 cGy/h). The mean number of TTT applications was three (range, 1–6), and the mean TTT power was 750.7 (range, 400–1000) mW.

The mean tumor thickness decreased from 5.2 mm (range, 2.0–8.0 mm) to 3.1 mm (range, 0–8.0 mm, a 39.3% decrease) at a mean follow-up of 44.9 months (range, 19.0–94.0). Mean tumor thickness was 4.3 mm at 3 months, 4.1 mm at 6 months, 3.5 mm at 12 months, 3.2 mm at 18 months, 2.7 mm at 24 months, 3.2 mm at 30 months, 2.6 mm at 36 months, 2.8 mm at 42 months, 2.5 mm at 48 months, 1.6 mm at 54 months, and 1.6 mm at 60 months (Fig. 3). The decrease in tumor thickness was significant compared to baseline at 6 months and thereafter (log-rank test, $P =$

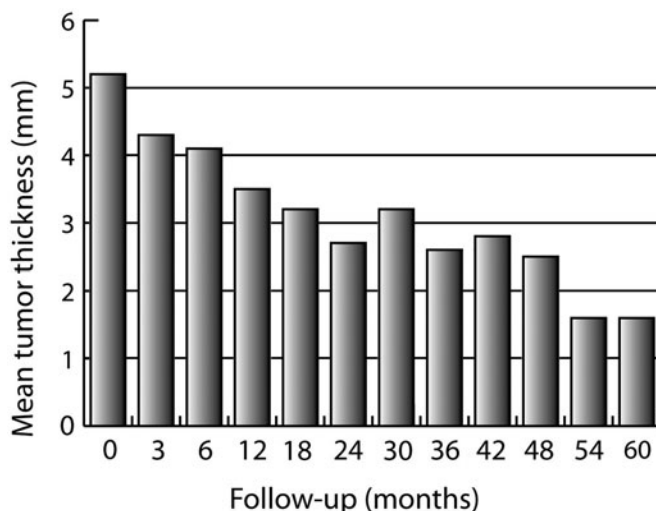


Figure 3. Mean tumor thickness decrease in patients treated with combined Ru-106 plaque radiotherapy and TTT (group B).

0.015). Kaplan-Meier curves predicted an at least 50% decrease in tumor thickness in 81% of the eyes in group B by 48 months after treatment (Fig. 2).

The regressed tumors became more reflective on A-mode ultrasonography and more acoustically solid on B-mode ultrasonography. On fluorescein angiography, tumors treated with Ru-106 plaque radiotherapy and TTT demonstrated hypofluorescence owing to the accumulation of melanophagic pigment over the tumor surface.

Complications in group B were retinopathy in 13 eyes (43.3%), pigment dispersion into the vitreous cavity in six eyes (20.0%), an increase in exudative retinal detachment in five eyes (16.7%), vitreous hemorrhage in three eyes (10.0%), cataract in two eyes (6.7%), and papillopathy in two eyes (6.7%). As determined by fluorescein angiographic studies, only one eye with retinopathy required laser photocoagulation. Phacoemulsification surgery and intraocular lens implantation were performed in three eyes. Vitreous hemorrhage was spontaneously resorbed in all eyes. Eyes with increase in exudative retinal detachment after Ru-106 plaque radiotherapy were treated with a tapering dose of 1 mg/kg per day of oral prednisolone with complete resolution of the exudative retinal detachment.

We compared the complication rates in 19 eyes with tumor thickness ≤ 5 mm with the complication rates in 11 eyes with tumors from >5 to <8 mm in thickness. The complications seen in eyes with tumors ≤ 5 mm in thickness were retinopathy in eight eyes (42.1%), pigment dispersion into the vitreous cavity in three eyes (15.8%), exudative retinal detachment in two eyes (10.5%), radiation papillopathy in two eyes (10.5%), cataract in two eyes (10.5%), and vitreous hemorrhage in one eye (5.3%). The complications seen in eyes with tumors from >5 to <8 mm in thickness were retinopathy in five eyes (45.5%), exudative retinal detachment in three eyes (27.3%), pigment dispersion into the vitreous cavity in three eyes (27.3%), and vitreous hemorrhage in two eyes (18.2%). Because of the small number of patients in each complication group, statistical analysis could not be performed. However, there seemed to be no obvious differences with respect to the complication rates between the two groups.

Globe conservation was achieved in 16 of 19 eyes (84.2%) with tumors ≤ 5 mm in thickness and in ten of 11 eyes (90.9%) with tumors from >5 to <8 mm in thickness. There was no significant difference between these groups with respect to globe salvage (Fisher exact test, $P > 0.05$).

Excluding the eyes that were enucleated, ten of 26 eyes (38.5%) had VA $\geq 20/40$ initially. At a mean follow-up of 44.9 months, only one of 26 eyes (3.8%) maintained a VA $\geq 20/40$. Sixteen of 26 eyes (61.5%) had VA $\leq 20/200$ at baseline, and 19 of 26 eyes (73.1%) had VA $\leq 20/200$ at the final follow-up (excluding eyes that underwent enucleation and exenteration). On the logMAR scale, VA decreased significantly ($P = 0.007$) from 0.9 at baseline to 1.3 at the final follow-up (Mann-Whitney U test).

Tumor recurred in five eyes (16.7%). Three eyes (10%) with intraocular tumor recurrence were enucleated. One eye (3.3%) with diffuse choroidal melanoma was exenter-

ated owing to the development of orbital melanoma in the absence of an intraocular tumor recurrence. This patient developed liver metastasis at the 36-month follow-up after plaque treatment.

Comparison of Groups A and B

There was no statistically significant difference ($P > 0.05$) between groups A and B with respect to patient age, tumor base diameter, tumor thickness, tumor distance to optic disc, tumor distance to foveola, or baseline VA (t test for independent samples for age, tumor base diameter, and tumor thickness; Mann-Whitney U test for tumor distance to optic disc and tumor distance to foveola; and Wilcoxon test for VA).

There was no statistically significant difference with respect to globe conservation rate between groups A and B (Fisher exact test, $P > 0.05$). There was no statistical difference in the rate of an at least 50% reduction in tumor thickness between groups A and B (log-rank test, $P > 0.05$). The difference between groups A and B with respect to radiation complications and metastasis could not be analyzed statistically because of the small number of patients.

Discussion

Our study showed no significant difference in the rate of globe conservation between group A (Ru-106 plaque radiotherapy) and group B (Ru-106 plaque radiotherapy combined with TTT). Therefore, adding TTT to Ru-106 plaque radiotherapy did not improve the globe conservation rate in choroidal melanomas <8 mm in thickness. Subgroup analysis showed that the globe conservation rate did not differ significantly between tumors ≤ 5 mm and tumors from >5 to <8 mm in thickness in each group.

Kaplan-Meier curves predicted at least a 50% decrease in tumor thickness in 92% of the eyes in group A within 4 years of treatment. The decrease in tumor thickness started to occur 6 months after treatment. Kaiserman et al.¹⁵ reported that at a mean follow-up of 2 years, tumor thickness had significantly decreased from 5.2 to 3.3 mm (36.5% decrease) in 147 patients with posterior uveal melanoma who received Ru-106 plaque radiotherapy. They found that posterior uveal melanomas regress, on average, to 60% of their initial thickness at 2 years after plaque radiotherapy.

In group B, Kaplan-Meier curves predicted an at least a 50% decrease in tumor thickness in 81% of the eyes within 4 years of treatment, numerically lower than the proportion (92%) in group A. However, the difference between groups A and B was not statistically significant. The decrease in tumor thickness started to occur 6 months after treatment, similar to group A. Bartlema et al.¹⁶ reported that after combined Ru-106 plaque radiotherapy and TTT, 45 of 50 patients (90%) with choroidal melanoma showed complete tumor regression (100% tumor thickness decrease) at the 5-year follow-up.

Treatment complications were similar in both groups. In our study, the most common complications seen in group A

were radiation retinopathy (20.8%), vitreous hemorrhage (20.8%), and cataract (8.3%). The relative proportion of radiation complications may be related to tumor location.^{17–20} Radiation cataract and neovascular glaucoma develop more often in anteriorly located ciliary bodies and choroidal melanomas after plaque radiotherapy, and radiation maculopathy is more frequent in posteriorly located tumors.^{17–20} There was a slightly increased risk of complications in tumors from >5 to <8 mm in thickness versus those ≤5 mm in thickness in group A, although the difference could not be studied statistically.

The most common complication seen in group B was retinopathy (43.3%), followed by pigment dispersion into the vitreous cavity (20%) and increase in exudative retinal detachment (16.7%). Shields et al.¹⁴ also found retinopathy to be the most common complication (39%) at 5 years after combined plaque radiotherapy and TTT.

Liver metastasis occurred in two patients in group A and one patient in group B. The difference in rates of complications and metastasis between groups A and B could not be analyzed statistically.

Concerning VA changes, there was a significant decrease in final VA compared to baseline VA in group B but not in group A. These results suggest that combining TTT with Ru-106 plaque radiotherapy may result in worse VA compared to monotherapy with Ru-106 plaque radiotherapy. However, group A patients had a markedly shorter follow-up compared to group B patients in our study, and with time, further vision loss may be expected in group A. When we compared some of the VA data to the literature, 14.3% of our patients had VA of 20/40 or better after Ru-106 plaque radiotherapy at a mean follow-up of 24.6 months (group A) versus 31% at the 5-year follow-up in the series of Bergman et al.⁹ and 55% at 9 years in the series of Damato et al.²¹ The poorer visual outcome with shorter follow-up may be related to the greater initial thickness of the tumors in our series. In our group B, the proportion of patients having VA ≤ 20/200 increased from 62% to 73% after treatment, compared to an increase from 10% to 46% in the series of Shields et al.¹⁴ The eyes in our series had worse baseline VAs than the patients in the series of Shields et al.,¹⁴ and following treatment, some of our patients had further vision loss caused by radiation and TTT complications.

Weaknesses of our study included the small number of patients and the relatively short follow-up period. Kaplan-Meier estimates were used to predict long-term regression rates longer than the mean follow-up. Despite these shortcomings, our study showed that compared to Ru-106 plaque radiotherapy alone, Ru-106 plaque radiotherapy combined with TTT did not result in a significant change in globe conservation rate or at least a 50% reduction in tumor thickness rate in choroidal melanomas <8 mm in thickness. Although Ru-106 plaque radiotherapy is mainly used for choroidal melanomas ≤5 mm in thickness, it can also be considered in selected tumors with thickness from >5 to <8 mm with comparable tumor control and slightly increased risk of radiation complications.

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